Sample Scenario

All of this is fictional and deliberately simplified. Where the spec is ambiguous, make a reasonable choice, flag it in your README, and move on. Spotting and resolving ambiguity is part of what we look at.

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1. Network

Meridian is a city served by five vertiports. Coordinates are in km on a flat plane; distances are straight-line and already rounded to 0.1 km.

ID	Name	x (km)	y (km)	Charging pads
V0	Central	0	0	3
V1	Airport	18	-6	2
V2	Harbour	-9	7	2
V3	TechPark	12	14	1
V4	NorthSuburbs	-4	24	1

Distance matrix (km)

	V0	V1	V2	V3	V4
V0	0.0	19.0	11.4	18.4	24.3
V1	19.0	0.0	30.0	20.9	37.2
V2	11.4	30.0	0.0	22.1	17.7
V3	18.4	20.9	22.1	0.0	18.9
V4	24.3	37.2	17.7	18.9	0.0

2. Time

- Time is in **integer minutes**. `t = 0` is 06:00. The operating day ends at `t = 360` (12:00); every flight must have landed and completed its post-landing turnaround by then.

3. Aircraft

- Fleet: 8 identical aircraft, A0–A7.
- Initial position (all parked, not on a pad, at `t = 0`): A0, A1 at V0; A2, A3 at V1; A4, A5 at V2; A6 at V3; A7 at V4.
- Initial state of charge (SoC): **90%** for all aircraft.
- One aircraft carries exactly one request per flight (no pooling).

4. Flight model

For a flight of distance `d` km:

- **Block time** (departure to landing): `4 + ceil(d / 3)` minutes. (4 min of terminal phases + cruise at 180 km/h.)
- **Energy**: `12 + 1.0 × d` percentage points of SoC. (Usable battery = 100 kWh, so 1% SoC = 1 kWh.)
- **Reserve**: SoC at landing must be **≥ 20%**. Equivalently, SoC at departure must be ≥ `20 + 12 + d`.

A flight is either a **revenue flight** (serves a request, origin/destination match the request) or a **deadhead** (repositioning with no passengers).

5. Ground operations: pads and charging

- Aircraft on the ground are either on a pad or parked. Parking is unlimited and has no chargers.
- A pad stint is a continuous interval during which an aircraft occupies one pad.
    - Every landing starts a pad stint. Every departure ends a pad stint.
    - Every pad stint lasts at least 5 minutes (turnaround / boarding / pre-flight).
    - A parked aircraft may start a new pad stint at any time (e.g. to charge, or to board).
    - An aircraft that lands and departs again without parking does so within a single stint (≥ 5 min).
- The number of simultaneous pad stints at a vertiport may not exceed its pad count. A flight cannot land if no pad is free at its landing time.

- Every pad has a charger. Charging only happens during pad stints. Charging rate:
  - 2.0 %/min while SoC < 80%
  - 2/3 %/min while 80% ≤ SoC < 100%
  - capped at 100%.
- A stint is the interval `[start, end)` in minutes. Charging is applied over its full length.

6. Demand

A **request** is a single booking from origin to destination with a departure window.

- Departure must occur in `[earliest_dep, latest_dep]` (window is 15 minutes).
- Requests become known 30 minutes before `earliest_dep` (requests with `earliest_dep < 30` are known at `t = 0`). The operator must accept or reject a request when it becomes known; an accepted request that is then not flown counts as a failure (see §7).
- Revenue: `40 + 4 × d` (already in the data).

Demand generator

The fixed instance in §8 is one sample from this generator (truncated to its first 70 requests). Candidates who need more episodes (e.g. for training or stochastic evaluation) should implement this generator.

- Arrivals of `earliest_dep` follow a non-homogeneous Poisson process on `[0, 300)` with rate
  `λ(t) = 0.2 × (1 + 1.4 × exp(−((t − 150) / 50)²))` requests/min (a morning peak around 08:30).
- `earliest_dep` = floor of the arrival time; `latest_dep = earliest_dep + 15`.
- Origin is drawn with weights `w = [1.6, 1.8, 1.0, 1.1, 0.7]` over V0–V4. Destination is drawn from the other four vertiports with the same weights, renormalised.

7. Objective and accounting

```
None
profit = Σ revenue of served requests
       − 1.50 × total km flown (revenue + deadhead)
       − 0.30 × total kWh consumed in flight
```

- Rejected requests earn nothing and cost nothing.
- An **accepted-then-failed** request is a service failure. Report it separately. Treat it as a hard error in any plan you claim is feasible.

- Also report: requests served, deadhead km, minimum SoC at any landing, pad utilisation per vertiport.

8. Reference baseline policy: JIT

Process requests in order of `earliest_dep` (which is also the order they become known). For each request:

1. For every aircraft, find the earliest feasible departure time `D` in the request window, given the aircraft's existing commitments. If the aircraft is not at the origin, include a deadhead to the origin first.
2. Just-in-time charging: before each departure, the aircraft takes one pad stint that is exactly as long as needed (minimum 5 min) to reach `20 + energy(next leg)`, ending at the departure time. After each landing it takes a 5-minute stint (charging during it), then parks.
3. Every stint must fit within pad capacity.
4. Among aircraft with a feasible plan, pick the one with the least deadhead km, breaking ties by earliest `D`, then lowest aircraft index. Accept and commit that plan. If no aircraft is feasible, reject.

JIT never repositions on its own initiative and never rejects a request it can serve.